Python Project - Week 4

File Processing - Save and Load

Today's goal

Use OPEN, READ, WRITE, and file conversion to save the current game state and restore it later.

1. Guided Coding

Study the example code below with your team. Type it into your editor and run it. Do not copy and paste.

```
name = "Alex"
hp = 85
gold = 50

with open("savegame.txt", "w") as fp:
    fp.write(name + "\n")
    fp.write(str(hp) + "\n")
    fp.write(str(gold) + "\n")

with open("savegame.txt", "r") as fp:
    loaded_name = fp.readline().strip()
    loaded_hp = int(fp.readline())
    loaded_gold = int(fp.readline())

print(loaded_name, loaded_hp, loaded_gold)
```

2. Team Coding Tasks

1. Create save_game(filename, player, inventory).

2. Save the player's name, HP, gold, and attack to savegame.txt.

3. Use a loop to save every inventory item.

4. Create load_game(filename) and restore the saved values.

5. Save a game, load it again, and confirm that the restored values match the original values.

3. Expected Output / Checkpoint

```
Game saved.

Loaded character:
name : Alex
hp : 85
gold : 50
attack : 15
Inventory: ['potion', 'slime jelly']
```

4. Submission Check

• The program runs without errors.

• The program runs from beginning to end.

• Every team member can explain the code added this week.

• Save the completed source code as week4_project.py.